

PLN x TerraMind

Demo Guide

*Satellite-Powered Infrastructure Intelligence
for Perusahaan Listrik Negara, Indonesia*

Powered by IBM TerraMind (ICCV 2025)
Any-to-Any Earth Observation Foundation Model

For presenters and non-technical stakeholders
Includes: Dashboard walkthrough, scenario objectives, full glossary

PLN x TerraMind -- Demo Guide

For presenters and non-technical stakeholders.

This guide walks you through every step of the demo, explains what is happening on screen in plain language, and includes a glossary so anyone can follow along.

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1. What This Demo Does

Imagine you have a single photograph taken by a satellite flying over a PLN power plant or transmission line. From that one photograph, the AI can generate four additional "views" that normally require completely different sensors, field surveys, or separate data purchases:

What the AI creates	What it tells PLN	Normally requires
Radar image (SAR)	What the area looks like through clouds and at night	A different satellite (Sentinel-1)
Elevation map (DEM)	How high or low the ground is (flood risk)	LiDAR survey or aerial mapping
Land cover map (LULC)	What is on the ground: trees, water, buildings, farms	Manual classification or separate dataset
Vegetation health (NDVI)	How thick and healthy the vegetation is	Specialized analysis of satellite bands

In short: 1 satellite photo in, 4 intelligence products out -- in seconds.

2. How to Start

Option A: Web Dashboard (recommended for presentations)

Open a terminal and run:

```
cd /Users/priyo/Downloads/terramind
source venv/bin/activate
streamlit run pln_demo/app.py
```

A browser window will open at <http://localhost:8501>.

Option B: Guided Script (automated slide-show style)

```
cd /Users/priyo/Downloads/terramind
source venv/bin/activate

# Quick run (~1 minute, lower quality -- good for testing)
python pln_demo/demo_script.py --quick

# Full quality (~10 minutes, best for presentations)
python pln_demo/demo_script.py

# Presenter mode -- pauses between each scenario for discussion
python pln_demo/demo_script.py --interactive
```

3. Dashboard Walkthrough (Step-by-Step)

Step 1: Open the Dashboard

After running `streamlit run pln_demo/app.py`, the browser shows the dashboard with 4 tabs at the top:

Tab	What it does
Overview	Shows PLN infrastructure map, site list, and use cases
Analysis	Run AI analysis on satellite images (the main event)
Generation	Free-form: pick any combination of outputs
About	Background information on the technology

Step 2: Look at the Sidebar (left panel)

The sidebar has the controls. Here is what each one means:

Model Settings

Control	What it means	What to choose
Model Size	How big and accurate the AI brain is. Bigger = better results but slower.	tiny for quick test, base for best results
Diffusion Timesteps	How many "refinement passes" the AI makes when creating an image. More passes = cleaner result, like an artist doing more drafts.	2 for fast preview, 10 for good quality, 30 for best
Device	What hardware runs the AI. Shown automatically.	cuda = NVIDIA GPU (fastest), mps = Apple chip (good), cpu = any computer (slowest)

Data Source

Control	What it means
Example Data	Pre-loaded satellite images that come with the demo
Upload GeoTIFF	Use your own satellite image file (must be Sentinel-2 format)
Load Example button	Click to load the selected sample image into the system

Initialize Model button

Click this to load the AI model into memory. This downloads the model weights the first time (~500 MB) and takes 10-30 seconds. You only need to do this once per session.

Step 3: Load Data + Initialize Model

1. In the sidebar, keep "Example Data" selected
2. Pick any file from the dropdown (e.g., 282D_485L_3_3.tif)
3. Click "Load Example"
4. Click "Initialize Model"
5. Wait for the green "Model loaded!" message

Step 4: Explore the Overview Tab

This tab shows:

- Top metrics: Number of monitored sites, total power capacity
- Site table: List of 9 PLN power plants with type, capacity, region
- Interactive map: Click markers to see plant details. Color-coded:
- Red = Coal, Blue = Hydro, Orange = Gas, Green = Geothermal
- Use cases: Expandable cards explaining each monitoring application

Presenter tip: Use the map to show the geographic spread across Java-Bali, Sumatra, Kalimantan, and Sulawesi.

Step 5: Run an Analysis

Switch to the "Analysis" tab. You will see:

1. A preview of the satellite image you loaded
2. A dropdown to pick which analysis to run
3. A big blue "Run Analysis" button

Available Analyses

Analysis	What it answers	What the AI generates
Vegetation Encroachment	"Are trees growing dangerously close to our power lines?"	Vegetation index + Land cover + Risk map
Flood Risk Assessment	"Could this substation flood during monsoon?"	Radar image + Elevation map + Flood susceptibility
Land Use Change	"Is urbanization expanding near our infrastructure?"	Land cover classification map
Full Multi-Modal Survey	"Give me everything about this site"	All 4 outputs at once

Reading the Results

After clicking "Run Analysis", you will see:

- Metric cards at the top: Key numbers summarizing the finding
- Alert banner: Color-coded assessment
 - Red = HIGH RISK, action needed
 - Yellow = MODERATE, schedule attention
 - Green = LOW RISK, area is fine
- Side-by-side images: Original photo + AI-generated maps
- Charts: Bar/pie charts showing land cover breakdown

Step 6: Try the Generation Tab

This is the "freestyle" mode. You can:

1. Pick any combination of output types (SAR, DEM, LULC, NDVI)
2. Click "Generate"
3. See the original image alongside all generated outputs

Presenter tip: This is the "wow moment" -- show how 1 image produces multiple completely different data products.

Step 7: Show the About Tab

Quick reference for stakeholders who want to understand the technology and see links to the research paper and model repository.

4. Command-Line Demo Walkthrough

The command-line demo (demo_script.py) runs all 5 scenarios automatically and saves images to pln_demo/demo_output/.

What Happens When You Run It

1. Banner appears: "PLN x TerraMind"

2. Shows configuration (model size, device)

3. Loads the AI model (one-time download)

4. Runs Scenario 1 -> Scenario 5 sequentially

5. Each scenario:

- Explains the PLN problem (narration text)

- States the objective

- Loads satellite image

- Runs AI inference

- Prints metrics with color-coded risk levels

- Saves comparison images to demo_output/

- Gives a recommendation

6. Final summary with value propositions and next steps

Command Options Explained

Command	What it does	When to use
--quick	Uses smallest model, fewest refinement steps	Testing, time-limited demos
--model-size base	Uses the best model	Formal presentations
--timesteps 10	10 refinement passes (good balance)	Default, recommended
--scenario 2	Run only scenario 2 (flood risk)	When audience cares about specific topic
--interactive	Pauses after each scenario, press Enter to continue	Live presentations with discussion

5. The 5 Demo Scenarios

Scenario 1: Vegetation Encroachment Risk

"Are trees threatening our 500kV transmission lines?"

The problem	Tropical trees grow fast. When branches touch 500kV lines, it causes electrical flashovers (like lightning), wildfires, or blackouts affecting millions.
Today's solution	Helicopter patrol at Rp 15-25 million/hour. Expensive, limited coverage, weather-dependent.
AI solution	Satellite scans the entire corridor. AI creates a vegetation health map (NDVI) and land cover map (LULC), then calculates a risk score for each area.
What you see	4 panels: satellite photo, vegetation map (green=dense), land type map, and a red/yellow/green risk map.
Key number	"Critical Risk %" -- the percentage of the corridor where tall trees or dense vegetation could reach power lines.
Action	If >30% critical: dispatch ground crew immediately. If 10-30%: schedule trimming. If <10%: corridor is clear.

Scenario 2: Flood Risk Assessment

"Could our coastal substation flood during monsoon season?"

The problem	Indonesia has 2,000+ flood events/year. PLN lost Rp 1.2 trillion to flood damage in 2024. Substations with expensive transformers sit in low-lying areas.
Today's solution	React after flooding occurs. Emergency repairs take weeks.
AI solution	From a satellite photo, AI generates a radar view (works through clouds) and an elevation map. Low-lying areas near water = high flood risk.
What you see	4 panels: satellite photo, radar image, elevation map (brown=high, green=low), flood susceptibility (blue=danger).
Key number	"Mean Elevation" -- if below 10 meters, the area is critically vulnerable. "High Flood Risk %" shows how much area could flood.
Action	If >50% high risk: install flood barriers, elevate transformers, stage mobile pumps BEFORE October monsoon.

Scenario 3: Urban Expansion & Load Forecasting

"Where is urbanization outpacing our grid capacity?"

The problem	Indonesia urbanizes at 4.1%/year. New housing and factories need power. PLN must build infrastructure BEFORE demand arrives, not after blackouts.
Today's solution	Census data (5-year lag) and manual surveys. Too slow.
AI solution	Satellite photo becomes a land cover map showing exactly what is on the ground: buildings, farms, forest, water. Track changes over time.
What you see	Satellite photo next to a color-coded land use map + bar chart showing percentages.
Key number	"Built Area %" -- high means dense urban (high electricity demand). "Agriculture %" converting to built = growth area needing new substations.
Action	Plan 150kV feeder line expansion and new substations in areas with >20% agricultural-to-urban conversion.

Scenario 4: Terrain & Route Planning

"What is the best route for our new 275kV transmission line?"

The problem	Building transmission lines through mountains requires knowing exact terrain. Ground surveys cost Rp 50-100 million/km and take 6+ months.
Today's solution	Slow, expensive field surveys before any route can be approved.
AI solution	From a satellite photo, AI generates an elevation map. Engineers can screen dozens of candidate routes in minutes instead of months.
What you see	Satellite photo, radar image, and elevation map with color gradient.
Key number	"Elevation Range" -- tells you how hilly the route is. Flat = easy, steep = expensive. "Elevation Variance" -- high variance means difficult tower placement.

Action

Route along ridges (avoid valleys that flood). Use AI elevation as preliminary screen, then ground-truth only the top 2-3 candidate routes.

Scenario 5: Comprehensive Multi-Modal Survey

"Give me the complete picture for our new substation site."

The problem	Siting a new Rp 200+ billion substation requires terrain, vegetation, land use, flood risk, and all-weather data. Procuring all this separately takes weeks and multiple vendors.
Today's solution	Order DEM from one vendor, SAR from another, land cover from a third. Wait weeks, pay \$\$\$.
AI solution	ONE satellite photo generates ALL FOUR data products in seconds. Complete site intelligence package.
What you see	5 panels side-by-side: original photo + 4 generated products (SAR, DEM, LULC, NDVI).
The "wow" moment	One input, four outputs, seconds of processing. This is what replaces weeks of multi-vendor procurement.
Action	Compare candidate sites, shortlist the best 2-3, then send a field team only to those locations.

6. Glossary -- Every Term Explained Simply

Satellite & Data Terms

Term	Simple Explanation	PLN Analogy
Sentinel-2	A free European satellite that takes photographs of Earth every 5 days. Like Google Earth but updated regularly with 12 color channels (not just RGB).	Like having a free helicopter with a very good camera flying over all of Indonesia every 5 days.
Sentinel-1	A different satellite that uses radar instead of a camera. Works through clouds and at night.	Like having a helicopter with night-vision that can see through monsoon clouds.
GeoTIFF	A standard file format for satellite images. Like a JPEG but with geographic coordinates embedded so you know exactly where the image is on Earth.	Like a labeled photograph that includes GPS coordinates for every pixel.
Bands / Spectral Bands	A satellite camera captures light in multiple wavelengths (colors), including infrared that human eyes cannot see. Sentinel-2 has 12 bands.	Like having 12 different colored flashlights -- each one reveals different things about the ground below.
Pixel	One tiny square in the satellite image. For Sentinel-2, each pixel covers 10m x 10m on the ground.	One pixel = one small room on a map.
224 x 224	The image size the AI expects: 224 pixels wide by 224 pixels tall. At 10m/pixel, this covers approximately 2.2 km x 2.2 km.	A square patch of land roughly 2 km on each side.

AI Output Terms (The 5 Modalities)

Term	Full Name	What It Shows	How to Read the Colors
S2 L2A	Sentinel-2 Level 2A	The original satellite photograph in true color	Normal photo: green=vegetation, blue=water, brown=soil, gray=buildings
SAR / S1 GRD	Synthetic Aperture Radar (Sentinel-1 Ground Range Detected)	Radar view of the same area. Works through clouds, day and night.	Bright = rough surfaces (buildings, forest). Dark = smooth surfaces (water, flat ground).
DEM	Digital Elevation Model	A height map showing how high or low the ground is	Brown/dark = high ground. Green/light = low ground. Used to find flood-prone valleys.
LULC	Land Use / Land Cover	A classified map showing WHAT is on the ground	Each color = a different land type (see table below). Like a zoning map.
NDVI	Normalized Difference Vegetation Index	A measure of how green and healthy vegetation is	Green = healthy dense vegetation. Yellow = sparse. Red/brown = bare soil or water. Values range -1 to +1. Above 0.5 = very dense vegetation.

LULC Color Classes (What Each Color Means)

Color	Class	Description	Risk for Power Lines
Black	No Data	No information	--
Blue	Water	Rivers, lakes, sea	Low (but flood concern)
Dark Green	Trees	Forest, tall trees	HIGH -- branches can touch lines
Light Green	Grass	Short grass, lawns	Low -- too short to reach lines
Yellow-Green	Flooded Vegetation	Swamp, mangrove, wetland	HIGH -- flooding + vegetation
Yellow	Crops	Agricultural fields	Medium -- seasonal tall crops
Purple	Scrub / Shrub	Bushes, low woody plants	Medium -- can grow into lines
Light Blue	Built Area	Buildings, roads, cities	Low -- managed clearance
Orange-Brown	Bare Ground	Exposed soil, rock	None -- clear corridor
White	Snow / Ice	(Not applicable in Indonesia)	--

AI Model Terms

Term	Simple Explanation	Analogy
TerraMind	The AI model created by IBM and the European Space Agency. It learned from millions of satellite images to understand Earth.	A very experienced satellite analyst who has studied millions of images and can "imagine" what the terrain, vegetation, and radar would look like just from a photograph.
Foundation Model	A large, general-purpose AI trained on massive data. Can be adapted to many tasks without starting from scratch.	Like hiring a senior engineer who already knows the fundamentals -- you just need to brief them on PLN-specific requirements.
Any-to-Any Generation	The AI can take any type of satellite data as input and generate any other type as output. Photo to radar, radar to elevation, etc.	Like a translator who can translate between any pair of languages, not just one specific pair.
Diffusion / Diffusion		

Steps /
Timesteps

The method the AI uses to create images. It starts with random noise (TV static) and gradually refines it into a clear image.

More steps = clearer result.

Like an artist starting with a rough sketch and adding detail with each pass. 2 passes = rough draft, 10 passes = good painting, 30 passes

= museum quality.

Tokenizer	A component that translates between satellite data and the AI's internal language. Each data type (SAR, DEM, LULC) has its own tokenizer.	Like a specialized interpreter -- one for radar language, one for elevation language, one for land cover language.
Inference	Running the AI model to get results. "Running inference" = asking the AI to analyze an image.	"Running the analysis" -- the moment the AI is working on your image.
Model Size (tiny/small/basic/large)	How many "brain cells" the AI has. Larger models make better predictions but need more computing power and time.	Like choosing between a junior analyst (tiny, fast), a senior analyst (base, recommended), or a whole department (large, best but slowest).
Fine-tuning	Teaching the pre-trained AI to be even better at a specific task using PLN's own labeled data.	Like a senior engineer doing on-the-job training specific to PLN's corridors and infrastructure.
Pretrained	The model already learned from millions of images before we use it. We don't train from zero.	The analyst already has 10 years of experience before joining PLN.

Analysis Metrics

Metric	What It Means	When to Worry
NDVI Mean	Average vegetation health across the image. Range -1 to +1.	Above 0.5 = very dense vegetation near lines
High Vegetation %	Percentage of area with NDVI > 0.5 (thick vegetation)	Above 30% in a transmission corridor
Critical Risk %	Percentage of area covered by trees or flooded vegetation	Above 20% = schedule clearing, above 40% = urgent
Elevation Min/Max	Lowest and highest points in the area (meters above sea level)	Min below 5m near coast = flood risk
High Flood Risk %	Percentage of area susceptible to flooding	Above 50% = install flood protection
Built Area %	Percentage covered by buildings and roads	High = high electricity demand zone

Infrastructure Terms

Term	Meaning
500kV / 275kV / 150kV	Voltage levels of transmission lines. Higher voltage = more power carried over longer distances. 500kV is the backbone (like a highway), 150kV is regional distribution (like a city road).
Substation (Gardu Induk)	A facility that transforms electricity between voltage levels. Contains expensive transformers vulnerable to flooding.
Transmission corridor	The strip of land (typically 40-60m wide) cleared beneath and around power lines. Vegetation must be kept short within this corridor.
Right-of-way (ROW)	The legal right PLN has to maintain clear space around transmission lines.
Flashover	An electrical arc that jumps from a power line to a nearby tree or object. Causes fires and blackouts.
PLTU / PLTA / PLTGU / PLTP	PLN power plant codes: U=Uap (steam/coal), A=Air (hydro), GU=Gas-Uap (combined cycle gas), P=Panas bumi (geothermal).

Computing Terms

Term	Meaning	For PLN
GPU (cuda)	Graphics Processing Unit -- specialized chip that makes AI run 10-50x faster.	Recommended for production use. NVIDIA cards work best.
MPS	Apple Silicon's GPU acceleration. Available on newer Macs.	Good for demos on MacBook.
CPU	Regular computer processor. AI runs but is much slower.	Usable for demos, not recommended for production.
Streamlit	A tool that creates web dashboards from Python code.	The technology behind the browser-based dashboard.
TerraTorch	IBM's toolkit for working with TerraMind models. Handles model loading, data processing, and training.	The "engine" that makes everything work behind the scenes.
HuggingFace	A website where AI models are stored and shared (like GitHub for AI). Models download from here on first use.	Where the AI brain is downloaded from. First run takes a few minutes.

7. Frequently Asked Questions

Q: Does PLN need to buy satellite data?

No. Sentinel-2 data is free, provided by the European Space Agency. New images are available every 5 days for any location on Earth.

Q: Does this work during monsoon / cloudy weather?

The input needs a cloud-free Sentinel-2 image. However, once processed, the AI generates a SAR radar view that represents all-weather visibility. For persistent cloud cover, actual Sentinel-1 SAR data can be used as input instead.

Q: How accurate are the AI-generated outputs?

The outputs are AI predictions, not ground truth. They are best used for screening and prioritization (e.g., which 10 corridors out of 1,000 need attention first). Critical decisions should be verified with field inspection. The demo includes "AI vs Ground Truth" comparison panels so you can see the quality yourself.

Q: Can this be customized for PLN specifically?

Yes. This demo uses the "out-of-the-box" model. With PLN's own labeled data (e.g., confirmed vegetation incidents, flood zones), the model can be fine-tuned to become much more accurate for Indonesian conditions. Fine-tuning configs are already provided in the configs/ folder.

Q: How large an area can be analyzed at once?

The standard processing covers ~2.2 km x 2.2 km per patch. For larger areas, the system uses "tiled inference" -- it processes the area in overlapping patches and stitches them together. The demo includes a

large-tile example covering 20 km x 20 km.

Q: What hardware does PLN need?

- For demos: Any modern laptop (Mac or PC)
- For production: A server with an NVIDIA GPU (e.g., A100, V100) to process thousands of images per day
- For cloud deployment: AWS, GCP, or Azure with GPU instances

Q: How fast is the analysis?

- GPU (NVIDIA A100): ~2-5 seconds per image
- Apple M-series: ~5-20 seconds per image
- CPU only: ~1-5 minutes per image

8. Technical Reference

Project Structure

```

terramind/
  pln_demo/
    app.py           # Streamlit web dashboard
    demo_script.py   # Guided 5-scenario CLI demo
    pln_analysis.py  # Core analysis engine
    pln_config.py    # PLN infrastructure data
    DEMO_GUIDE.md    # This guide
    demo_output/     # Generated visualizations
    output/          # CLI demo outputs
  demo_data/
    scenario_1..5/   # Organized example data per scenario
  configs/
    pln_vegetation_monitoring.yaml
    pln_flood_detection.yaml
    pln_land_use_classification.yaml
  examples/          # Raw TerraMesh example tiles
  notebooks/         # Original TerraMind Jupyter notebooks
  venv/              # Python virtual environment

```

Setup from Scratch

```

# 1. Clone repository
git clone https://github.com/IBM/terramind.git
cd terramind

# 2. Create Python 3.12 environment
python3.12 -m venv venv
source venv/bin/activate

# 3. Install dependencies
pip install terratorch==1.1 diffusers==0.30.0

```

```

pip install streamlit folium streamlit-folium rioxarray matplotlib

# 4. Run dashboard
streamlit run pln_demo/app.py

# 5. Run CLI demo
python pln_demo/demo_script.py --quick

```

Fine-Tuning with PLN Data

```

terratorch fit -c configs/pln_vegetation_monitoring.yaml
terratorch fit -c configs/pln_flood_detection.yaml
terratorch fit -c configs/pln_land_use_classification.yaml

```

Demo Output Files

All 19 output images are saved to pln_demo/demo_output/:

File	Content
s1a_coastal_vegetation_risk.png	4-panel vegetation analysis
s1a_validation.png	AI vs ground truth comparison
s1b_dense_vegetation_risk.png	Dense vegetation HIGH RISK
s2a_coastal_flood.png	Coastal flood assessment
s2a_validation.png	AI vs ground truth DEM/SAR
s2b_river_basin_flood.png	River basin flood assessment
s3a_jakarta_urban.png	Jakarta urban classification
s3b_periurban.png	Peri-urban transition zone
s4a_highland_terrain.png	Highland terrain analysis
s4a_dem_validation.png	AI vs ground truth elevation
s4b_mixed_terrain.png	Mixed terrain routing
s5_*_full_survey.png	5-panel comprehensive surveys

Demo Data Reference

```

demo_data/
  scenario_1_vegetation_risk/
    input/
      coastal_transmission_corridor.tif  # Coastal, 37% trees
      dense_vegetation_corridor.tif      # 88% flooded vegetation
    ground_truth/                        # Validation data

  scenario_2_flood_assessment/
    input/
      coastal_substation_area.tif        # 0-16m elevation
      river_basin_plant.tif              # 100-250m river basin

```

```

scenario_3_urban_expansion/
  input/
    jakarta_industrial_zone.tif      # 100% built area
    peri_urban_development.tif      # Mixed development

scenario_4_terrain_routing/
  input/
    highland_transmission_route.tif  # 1,200m plateau
    mixed_terrain_corridor.tif      # 100-250m varied

scenario_5_comprehensive_survey/
  input/
    5 example tiles for full generation

```

Closing Talking Points for Presentations

#	Value Proposition	Business Impact
1	Replace helicopter patrols with satellite AI	Save Rp 15-25M/hr x thousands of hours/year
2	5 intelligence products from 1 free satellite image	Eliminate multi-vendor data procurement costs
3	Screen transmission routes in seconds	Reduce 6-month survey process to days
4	All-weather radar generation	Maintain monitoring during Oct-Mar monsoon
5	Predictive flood assessment	Reduce portion of Rp 1.2T/year flood damage

Proposed Next Steps

1. Pilot: Select 1 transmission corridor for 6-month validation
2. Fine-tune: Label 500-1,000 PLN-specific training samples
3. Integrate: Connect to PLN GIS platform (ArcGIS/QGIS)
4. Deploy: Automated monitoring pipeline on PLN cloud
5. Expand: IKN (new capital) development monitoring